

Screening Test—Junior Level-1992

ITMA-92

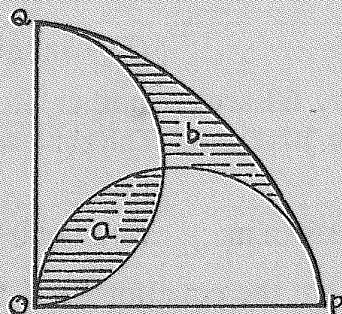
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1. The sum of two numbers is 2000 and their L.C.M. is 21879. The two numbers are
 (A) 1993, 7 (B) 1991, 9 (C) 1989, 11 (D) 1987, 13
2. If G.C.D. $(a, b) = 1$, then G.C.D. $(a + b, a - b)$ is
 (A) 1 (B) either 1 or 2 (C) either 1 or 3 (D) either 2 or 3.
3. a, b, c are integers such that G.C.D. $(a, b) = \text{G.C.D.}(a, b, c)$. The number of pairs of integers (x, y) such that $ax + by + c = 0$ is
 (A) 1 (B) 0 (C) finite (D) infinite.
4. If the last digit of $1^2 + 2^2 + 3^2 + \dots + n^2$ is 4 then the last digit of $1^{10} + 2^{10} + 3^{10} + \dots + n^{10}$ is
 (A) 0 (B) 4 (C) 3 (D) cannot be determined
5. a, b, c, d, e, f and t denote distinct digits and

$$\begin{array}{r}
 a \ b \ c \ d \ e \ f \times \\
 f \\
 \hline
 t \ t \ t \ t \ t \ t \\
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 \end{array}$$

The value of t is

- (A) 7 (B) 5 (C) 9 (D) 1
6. ABC is a rightangled triangle with $\angle B = 90^\circ$. M is the midpoint of AC and $BM = \sqrt{117}$. Sum of the lengths of the other two sides AB and BC is 30. The area of the triangle is
 (A) 108 (B) 27 (C) 112 (D) 110.
 7. OPQ is a quadrant of a circle and semicircles are drawn on OP and OQ . Areas a and b are shaded. a/b is

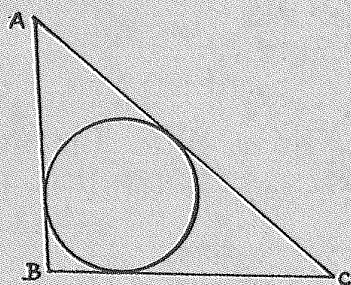


- (A) $\frac{1}{\sqrt{2}}$
 (B) $\frac{1}{2}$
 (C) $\frac{\pi}{4}$
 (D) 1

8. A circle is inscribed in an equilateral triangle. If the circumference has length 3 cms, then the perimeter of the triangle in cms is

- (A) $\frac{18}{\pi}$ (B) $\frac{9\pi\sqrt{3}}{2}$ (C) $\frac{9\sqrt{3}}{\pi}$ (D) 6

9. ABC is a triangle with sides 5 cms, 12 cms and 13 cms. A circle is inscribed in the triangle. The area of the circle in sq. cms is



(A) 4π

(B) 4

(C) π

(D) $\frac{6\pi}{13}$

10. A right angled triangle has hypotenuse p cms and one other side of length q cms. If $p - q = 1$, then the length of the remaining side is
- (A) 1 (B) $\sqrt{p+q}$ (C) $\sqrt{p^2+q^2}$ (D) $p^2 - q^2$
11. A ball of diameter 13 cms is floating so that the top of the ball is 4 cms above the smooth surface of a pond. What is the circumference, in cms, of the circle formed by the contact of the water surface with the ball?
- (A) 12π (B) 18π (C) 6π (D) $4\sqrt{22}\pi$
12. If $x + y = 1$ and $x^2 + y^2 = 2$, then $x^4 + y^4$ is
- (A) 4 (B) 8 (C) $3\frac{1}{2}$ (D) 3
13. If $v = \sqrt{2GM(\frac{1}{r} - \frac{1}{R})}$ then r equals
- (A) $\frac{2GMR}{v^2R - 2GM}$ (B) $\frac{2GMR}{v^2R + 2GM}$ (C) $\frac{2GM}{v^2R - 2GM}$ (D) $\frac{2GM}{v^2 - 2GMR}$
14. If $\frac{\sqrt{x+1} + \sqrt{x-1}}{\sqrt{x+1} - \sqrt{x-1}} = 3$, then x is
- (A) 3 (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{5}{3}$
15. Two vertical poles 20 m and 80 m high stand apart on a horizontal plane. The height in meters of the point of the intersection of the lines joining the top of each pole to the foot of the other, in metres, is
- (A) 18 (B) 50 (C) 16 (D) cannot be determined.
16. One electronic device makes a "beep" every 60 sec. Another electronic device makes a "beep" every 62 sec. They beeped together at 10:00 a.m. The time when they will next make a "beep" together is
- (A) 10:30 a.m. (B) 10:31 a.m. (C) 10:59 a.m. (D) 11:00 a.m.
17. The number of solutions in positive integers of the equation $3x + 5y = 1008$ is
- (A) 1 (B) 134 (C) 68 (D) 67
18. $|x + 1| + 2|x - 2| < 6$ is equivalent to
- (A) $-1 < x < 2$ (B) $0 < x < 1$ (C) $-1 < x < 3$ (D) $x < 2$

19. The area of the circle circumscribed about a regular hexagon is 2π . The area of the hexagon is

- (A) 6 (B) $3\sqrt{3}$ (C) $\sqrt{3}/2$ (D) $\sqrt{3}$

20. If $\log_{12} 27 = a$, then $\log_6 16$ is

- (A) $\frac{4(3-a)}{3+a}$ (B) $\frac{4(3+a)}{3-a}$ (C) $\frac{3+a}{4(3-a)}$ (D) $\frac{3-a}{4(3+a)}$

21. If $30x0103$ is a multiple of 13 then x is

- (A) 7 (B) 1 (C) 8 (D) 6

22. A man 1.5 m tall walks around the earth along a latitude circle. How much further does his head travel than his feet?

- (A) 3π m (B) 3 m (C) 64π m (D) π m

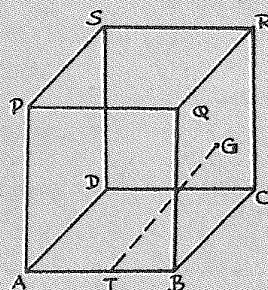
23. If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y, z are positive integers, then (x, y, z) is

- (A) (1, 5, 2) (B) (11, 2, 5) (C) (5, 2, 11) (D) (1, 2, 5)

24. If x and y are positive integers such that $3x + 7y$ is a multiple of 11, then which of the following must also be divisible by 11?

- (A) $4x + 6y$ (B) $x + y + 5$ (C) $9x + 4y$ (D) $4x - 9y$

25. $ABCDPQRS$ is a cube with side 10 cms. G is the centre of the square $BCRQ$ and T is the midpoint of AB . The length GT in cms is



(A) $5\sqrt{3}$

(B) 15

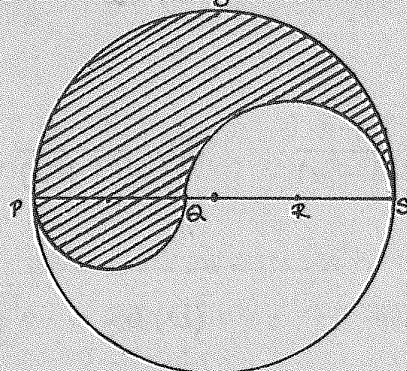
(C) 10

(D) $10\sqrt{5}$

26. A rhombus has half the area of the square with the same side length. The ratio of the long diagonal to the short one is

- (A) $2 + \sqrt{3}$ (B) $2 - \sqrt{3}$ (C) $1/2$ (D) 2

27. $PQRS$ is a diameter of a circle whose radius is r . The lengths PQ, QR, RS are equal. Semicircles are drawn on PQ and QS to create the shaded figure shown in the diagram. The perimeter of the shaded figure is

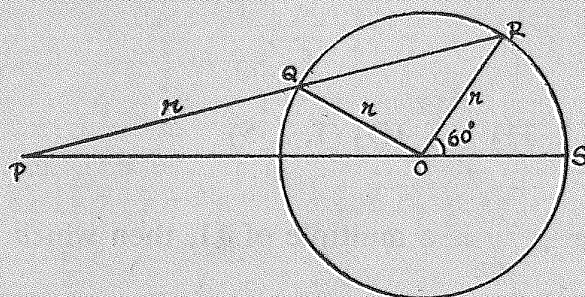


- (A) $2\pi r$
 (B) $\frac{4\pi r}{3}$
 (C) $\frac{5\pi r}{3}$
 (D) $\frac{3\pi r}{2}$

28. The length of one pair of opposite sides of a square is reduced by 10% and that of the other pair of sides is increased by 10%. The area of the new rectangle compared with that of the original square is

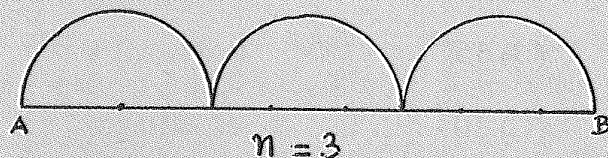
- (A) the same (B) 1% greater (C) 1% less (D) 5% greater

29. In the diagram, POS is a line through the centre O of a circle of radius r . The line PQR is drawn so that $PQ = r$. If $\angle ROS = 60^\circ$, then $\angle OPQ$ is



- (A) 10°
 (B) 15°
 (C) 20°
 (D) 25°

30. A line segment $AB = d$ is divided into n equal parts and semicircles are described on each part. The ratio of the area enclosed between the semicircles and the line AB to the perimeter of the figure obtained is



- (A) $\frac{\pi d}{4n(\pi + 2)}$
 (B) $\frac{d}{4n}$
 (C) $\frac{d}{4\pi n}$
 (D) $\frac{n\pi}{4}d$